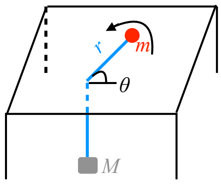
# Classical Mechanics - Home Assignment 3

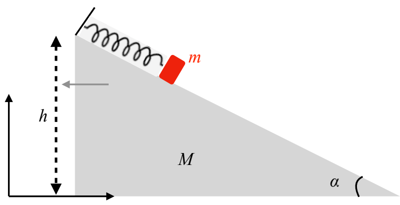
**Small Oscillations (September 2024)**

1. A mass $m$ is free to slide on a frictionless table and is connected, via a string that passes through hole in the table, to a mass $M$ that hangs below. Assume that $M$ moves in a vertical line only and that the string always remains taut. (a) Under what condition does $m$ undergo a purely circular motion? (b) When there is a small oscillation about this circular motion, what is the frequency of oscillation?




2. A (thin uniform) rod of length $\ell$ and mass $m$ is suspended at one end by a string of length $\ell$ from a fixed support under gravity. Find the frequency of small oscillations of the rod, assuming that the rod and the string lie in the same vertical plane.

3. A block of mass $m$ is attached to a wedge of mass $M$ by a spring with spring constant $k$.The block can move without friction on the inclined surface of the edge. The wedge in turn, is free to slide on a horizontal surface. Find the natural frequency of small oscillation of the block.




4. One pendulum is hung from the bob of another pendulum. The bobs of both have mass $m$ and the strings of both have length $l$. For oscillations confined to a vertical plane, find the frequencies of the normal modes of vibration.

5. Three point-masses $m$, $M$, $m$ are constrained to move on a horizontal circular groove of radius $a$. Each mass is connected by springs of spring constant $k$ to the other two and the springs themselves are straight and do not lie on the groove. In equilibrium, the masses are equidistant from each other and the springs have equal length. For small oscillations, find the frequencies of the normal modes of vibration.

6. Two pendulums of equal length $l$ and equal mass $m$ are hung from a horizontal support. The two points of suspension are at a distance $d$ apart. A spring of natural length $d$ and spring constant $k$ connects the two bobs. For oscillation in a vertical plane, find the frequencies of the normal modes of vibration.

7. A block of mass $M$ is attached to a horizontal spring of spring constant $k$ and is free to move on a horizontal plane. From this block is hung a simple pendulum consisting of a bob of mass $m$ and a string of length $\ell$. Find the frequencies of small oscillation of the bob. Assume that the spring and the pendulum lie in the same vertical plane.